

Slide 0 - Title

Today we'll walk you through our project "Computer Vision - Multi-Label Image Classification: Query2Label and MLSPL". I'm Christine, presenting with Andreas and Riajul. We will go into why multi-label learning matters, outline two complementary methods - Query2Label for feature recognition from the paper "Query2Label: A Simple Transformer Way to Multi-Label Classification", and ROLE loss for sparse annotations from the paper "Multi-Label Learning from Single Positive Labels", and describe how we reproduced them on MS-COCO 2014 benchmark dataset, and summarise the results.

Background: Why loss functions matter

Multi-label tasks differ from classic single-label classification because every image can contain several objects at once. That complicates training: the loss function must guide the network while coping with severe class imbalance and with labels that may simply be missing. Choosing that loss wisely is therefore crucial.

Losses covered

In this study we focus on four families of losses. Binary Cross-Entropy is the standard baseline. Expected Positive Regularisation tries to nudge the model so that each image has roughly the expected number of positives. ROLE—Regularised Online Label Estimation—alternates between updating the classifier and softly imputing missing labels. Finally, Focal and its Asymmetric variant down-weight easy negatives so that scarce positives get a stronger voice.

Binary Cross-Entropy refresher

BCE treats each class independently: if the dog label is present we penalise the log probability of "dog"; if not, we penalise the log probability of its complement. But that independence causes two issues. First, because positives are rare, negatives dominate the gradient; and second, any label that is simply missing is interpreted as a definite "no", inserting false negatives.

BCE limitations and Positive-only BCE

To mitigate the false-negative problem, practitioners sometimes discard all unobserved negatives and train only on positives—positive-only BCE. But that leaves the model with no incentive to predict zeroes, so we need extra regularisation, which brings us to EPR.

Expected Positive Regularisation

EPR keeps the positive-only loss yet adds a batch-level penalty: we tell the network the domain prior κ , roughly how many labels an image should contain, and we punish the squared distance between κ and the average number its own predictions switch on. That

soft constraint pulls the model back to reality without explicitly marking missing labels as negative.

ROLE

ROLE goes one step further. Alongside the classifier f it trains a label estimator g . Alternating optimisation lets g fill in fractional labels between 0 and 1, while f learns from those estimates plus the EPR regulariser. Intuitively, the two networks bootstrap each other, slowly recovering unobserved positives.

Experimental setup

Let me switch to implementation details. We reproduced Q2L and MLSPL on two RTX 3060 GPUs. For Q2L we matched the authors’ environment: Python 3.7, PyTorch 1.9 and minor patches such as rebuilding inplace-ABN and importing RandAugment. For MLSPL, which is lighter, we used Python 3.11 and PyTorch 2.2 on a single GPU to guarantee repeatability.

Dataset and metric

All experiments run on the MS-COCO 2014 benchmark—82 thousand training and 40 thousand validation images across 80 categories, each image resized to 448 or 576 pixels. Performance is reported in mean Average Precision, or mAP, the standard metric for multi-label classification.

Q2L backbones

We evaluated six backbones inside the Q2L framework: ResNet-101 at resolutions 448 and 576, TResNet-L with and without ImageNet-22k pre-training, and two transformer-style models: Swin-Large and CvT-w24. Memory overflow forced us to omit Swin and CvT from scratch, so we fine-tuned their 22k-pre-trained weights.

MLSPL protocol

For MLSPL we follow the paper exactly. Each image is reduced to a single positive label. We first train a linear classifier on frozen features for 25 epochs, grid-search learning rate and batch size, and then fine-tune end-to-end for another 10 epochs with a smaller learning rate.

Q2L results

Here are the numbers. Our Q2L reproductions line up almost perfectly with the paper: 84.9 mAP on ResNet-101 at 448 pixels and up to 91.3 mAP with the CvT backbone. That validates both the original claims and our environment.

MLSPL results

For MLSPL, the linear variant hits 66.3 mAP—identical to the paper—and fine-tuning nudges it to 66.9, slightly surpassing the reported 66.3. So even when each training image has only a single confirmed label, ROLE recovers roughly two-thirds of fully supervised performance on COCO.